

Homework (Carolina Reaper)

- Q1.** A guitarist's strumming speed increases by $3 \cdot 10^2$ beats per minute for every 5 minutes of practice. What is the slope of the line representing this situation?
- (A) 600
(B) 60
(C) $3 \cdot 10^2$
(D) 3
(E) 0
- Q2.** The frequency of a sound wave is directly proportional to the slope of its graph. If the slope is -2 , what is the meaning of this slope in the context of sound waves?
- (A) increasing pitch
(B) decreasing pitch
(C) constant pitch
(D) no pitch
(E) undefined pitch
- Q3.** An artist mixes 3 parts blue paint with 5 parts red paint to create a shade of purple. If x represents the amount of blue paint and y represents the amount of red paint, what is the slope of the line representing the ratio of blue to red paint?
- (A) $-\frac{5}{3}$
(B) $-\frac{3}{5}$
(C) $\frac{3}{5}$
(D) $\frac{5}{3}$
(E) 0
- Q4.** A rhythm pattern has a rate of change of 4 beats per measure. If the pattern starts with 2 beats, what is the equation of the line representing this situation?
- (A) $y = 4x + 2$
(B) $y = 4x - 2$
(C) $y = 2x + 4$
(D) $y = 2x - 4$
(E) $y = -4x + 2$
- Q5.** The table below represents the relationship between the size of a canvas and the number of brushstrokes required to paint it. What is the slope of the line representing this situation?
- (A) 2
(B) -2
(C) $\frac{1}{2}$
(D) $-\frac{1}{2}$
(E) 0
- Q6.** A musician's income is directly proportional to the number of concerts performed. If the slope of the line representing this situation is 1500, what does this slope represent?
- (A) \$1500 per concert
(B) \$1500 per hour
(C) 1500 concerts per year
(D) 1500 dollars per musician
(E) 1500 musicians per concert
- Q7.** The graph below represents the relationship between the frequency of a sound wave and its wavelength. What is the slope of the line representing this situation?
- (A) -1
(B) -2
(C) 1
(D) 2
(E) 0
- Q8.** An equation of a line is given by $y = -3x + 2$. What is the meaning of the slope in this equation in the context of a musical composition?
- (A) increasing tempo
(B) decreasing tempo
(C) constant tempo
(D) no tempo
(E) undefined tempo

Open Ended Questions (FRQ)

- Q9.** A painter's fee is directly proportional to the size of the canvas. If the fee is \$200 for a 4-foot by 6-foot canvas, what is the equation of the line representing the relationship between the size of the canvas and the fee? Show all work and explain your answer.
- Q10.** A music teacher notices that the number of students enrolled in her classes is decreasing by 2 students per week. If there are currently 15 students enrolled, what is the equation of the line representing the relationship between the number of weeks and the number of students enrolled? Show all work and explain your answer.

Chef's Tips

- Tip 1: Ensure students understand the concept of slope and its relationship to rate of change.
- Tip 2: Emphasize the importance of interpreting slope in different contexts, such as arts and music.
- Tip 3: Encourage students to show all work and explain their answers for free-response questions.